ALLAINGAYE LUCIEN
FULL-STACK SOFTWARE ENGINEER
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PROFESSIONAL SUMMARY

Junior Full-Stack Software Engineer with a B.Sc. in Software Engineering and hands-on experience building AI applications, fintech workflows, and real-time collaboration platforms. Develops secure systems using Python, FastAPI, React, PostgreSQL, Redis, and Docker, with a focus on payment safeguards, API reliability, performance optimization, and automated testing.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, PHP, Java, SQL
Backend: FastAPI, Django/DRF, Django Channels, Spring Boot, Laravel, Node.js
Frontend: React 18, Vite, Tailwind CSS, AG Grid, Recharts, Framer Motion, Axios
Data: PostgreSQL, Redis, SQLite, MySQL, SQLAlchemy, Hibernate
AI / LLM: OpenAI, Groq, OpenRouter, Gemini, DeepSeek, multi-provider fallback
Security: ECDSA P-256 (WebCrypto), idempotency keys, JWT, bcrypt, RBAC, CSP/HSTS
APIs & Methods: REST, WebSockets, Webhooks, Postman, Swagger, CORS, Agile, code review, collaboration, data structures and algorithms
DevOps & Testing: Docker, Docker Compose, GitHub Actions CI/CD, Nginx, Git, pytest, Playwright, Pest, Vitest

WORK EXPERIENCE

Software Engineering Intern | TechSpire Ltd · May 2025 – July 2025

- Built and documented 15+ REST API endpoints with Spring Boot and PostgreSQL for an e-commerce platform, supporting reliable order workflows.
- Integrated secure payment gateways and deterministic order-processing flows, contributing to a reported 99.9% transaction reliability rate.

FEATURED PROJECTS

PayPilot Shield AI Agent for Supplier Invoice Payments.

PayPal AI Hackathon (Team of 2) Live demo **:** paypilot.donambi.dev

- Architected a secure fintech agent where the AI model only proposes payment actions, while deterministic Python code enforces strict validation rules, ensuring zero direct LLM control over financial assets.
- Built client-side browser verification requiring owners to approve over-limit expenditures via non-extractable WebCrypto ECDSA P-256 keys.
- Prevented double-payment vulnerabilities using custom distributed idempotency keys (pp{request id}-{created_at}) integrated with PostgreSQL advisory locks to block parallel racing conditions.
- Implemented a zero-dependency verification module that re-checks policy, budget limits, snapshots, and replay guards immediately prior to invoking external APIs

Resume Analyzer AI-Powered Resume Optimization Platform.

Open-source project · React, FastAPI, PostgreSQL, Redis, Docker

- Increased user engagement by 40% through a multi-provider LLM fallback path across OpenRouter, Groq, Gemini, and DeepSeek.
- Reduced resume-analysis time by 60% to under 3 seconds through Redis caching and database-query optimization.
- Delivered PDF/DOCX/TXT parsing, JWT/bcrypt authentication, resume version tracking, job-description comparison, and AI-generated career content.

EDUCATION

Bachelor of Science (B.Sc.) in Software Engineering | Adventist University of Central Africa (AUCA)

CERTIFICATIONS & LANGUAGES

Certifications: AWS Certified Cloud Practitioner Pathway • Cybersecurity Fundamentals • AI Prompt Engineering • Advanced Network Operations 2.0v (Internet Society) • Cisco Networking Academy Essentials • JavaScript Essentials 2 • Introduction to Python (Sololearn)
Languages: English (Fluent) • French (Fluent)